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|                                                                                                                                                                                                                   |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |                                   |  |               |  |                           |  |              |  |                   |  |
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| F                                                                                                                                                                                                                 |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |                                   |  |               |  |                           |  |              |  |                   |  |
| <div>UniSec</div> <div>Medium Voltage Switchgear</div> <div>PT MULTIMAS NABATI ASAHAN</div> <div>MVD-20 &amp; MVD-21 20kV Boiler Plant</div> <div>SCHEMATIC DIAGRAM</div> <div>SBS-W BUS TIE</div> <div>H05</div> |  |                     |  |            |  |            |  |                                    |  |    |  |                        |  |                                   |  |               |  |                           |  |              |  |                   |  |
|                                                                                                                                                                                                                   |  |                     |  | Date Prep. |  | 15.11.2018 |  | PROJECT TITLE                      |  |    |  | ABB                    |  | EPMV                              |  | COVER SHEET   |  | Scale                     |  | NTS          |  | = H05             |  |
|                                                                                                                                                                                                                   |  |                     |  | Prepared   |  | YR         |  | MV Switchgear for MNA Serang Plant |  |    |  | PT. ABB Sakti Industri |  | MVD-20 & MVD-21 20kV Boiler Plant |  | Customer      |  | PT Multimas Nabati Asahan |  | & 307        |  |                   |  |
| 00                                                                                                                                                                                                                |  | Issued for Approval |  | 11.01.19   |  | YR         |  |                                    |  |    |  |                        |  |                                   |  | Checked       |  | AW                        |  | Drawing Nbr. |  | 18.042-MNA-307.03 |  |
| Ind.                                                                                                                                                                                                              |  | Revision            |  | Date       |  | Name       |  | Approved                           |  | AI |  | Based on: BE318042     |  |                                   |  | SBS-W BUS TIE |  | Customer Doc. Nbr.        |  |              |  | Cont. 27          |  |
| 1                                                                                                                                                                                                                 |  | 2                   |  | 3          |  | 4          |  | 5                                  |  | 6  |  | 7                      |  | 8                                 |  |               |  |                           |  |              |  |                   |  |

UniSec

Medium Voltage Switchgear  
PT MULTIMAS NABATI ASAHAN  
MVD-20 & MVD-21 20kV Boiler Plant  
SCHEMATIC DIAGRAM  
SBS-W BUS TIE  
H05

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# SHEET INDEX

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|      |                     |          | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                    | EPMV | Index of Sheet<br>MVD-20 & MVD-21 20kV Boiler Plant<br>SCHEMATIC DIAGRAM<br>SBS-W BUS TIE | Scale              | NTS          | = H05             |
|      |                     |          | Prepared   | YR         |                                                         |                        |      |                                                                                           |                    |              | Customer          |
| 00   | Issued for Approval | 11.01.19 | YR         | Checked    |                                                         | AW                     |      |                                                                                           |                    | Drawing Nbr. | 18.042-MNA-307.03 |
| Ind. | Revision            | Date     | Name       | Approved   | AI                                                      | PT. ABB Sakti Industri |      |                                                                                           | Customer Doc. Nbr. |              | Cont. 27          |
|      |                     |          |            |            | Based on:                                               | BE318042               |      |                                                                                           |                    |              |                   |

(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                                     |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DESIGNATION         |                                                                                                                                                     |      |            | DESCRIPTION |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -AA                 | MULTIFUNCTION UNIT (CENTRAL UNIT)                                                                                                                   |      |            |             |                                    |          |               |                        |                                                             |                        | -BGB8...-BGB10            | POSITION SWITCHES OF CIRCUIT-BREAKER                                                                                                |                   |       | B |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAD1,-BAD4         | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1                                                                                                      |      |            |             |                                    |          |               |                        |                                                             |                        | -BGB11                    | POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAD2,-BAD5         | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2                                                                                                      |      |            |             |                                    |          |               |                        |                                                             |                        | -BGB12                    | PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION                                                                        |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAD3,-BAD6         | CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3                                                                                                      |      |            |             |                                    |          |               |                        |                                                             |                        | -BGB13                    | PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION                                                                      |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAR                | VOLTAGE PROTECTION RELAY                                                                                                                            |      |            |             |                                    |          |               |                        |                                                             |                        | -BGD                      | POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.                                                                                |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAS1               | VOLTAGE SENSOR LOCATED ON PHASE L1                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGD1                     | POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR                                                                                 |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAS2               | VOLTAGE SENSOR LOCATED ON PHASE L2                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGD2                     | POSITION SWITCH OF CABLE COMPARTMENT DOOR                                                                                           |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BAT1               | VOLTAGE TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGE2                     | POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION                                                                |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCP                | TRANSFORMER PROTECTION RELAY                                                                                                                        |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI1                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCR                | CURRENT PROTECTION RELAY                                                                                                                            |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI2                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION                                                                   |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCS1               | CURRENT SENSOR LOCATED ON PHASE L1                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI3                     | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                 |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCS2               | CURRENT SENSOR LOCATED ON PHASE L2                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI4                     | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCS3               | CURRENT SENSOR LOCATED ON PHASE L3                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI5                     | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCT1               | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI6                     | POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION                             |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCT2               | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI7                     | POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION                                        |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCT3               | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI8                     | POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS                                                                                       |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -BCT4,-BCT7         | CURRENT TRANSFORMER LOCATED ON PHASE L1                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI9                     | POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR                                               |                   |       | E |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCT5,-BCT8         | CURRENT TRANSFORMER LOCATED ON PHASE L2                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI11                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION                                                                    |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCT6,-BCT9         | CURRENT TRANSFORMER LOCATED ON PHASE L3                                                                                                             |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI12                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION                                                                  |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BCZ                | DISTANCE PROTECTION RELAY                                                                                                                           |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI13                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BEF                | FREQUENCY PROTECTION RELAY                                                                                                                          |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI14, -BGI15            | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BER                | SUPERVISION RELAYS                                                                                                                                  |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI21                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION                                                                    |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BES                | SYNCHRONIZING RELAY                                                                                                                                 |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI22                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION                                                                  |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BET                | THERMAL PROTECTION RELAY                                                                                                                            |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI23                    | POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)                |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BGA1               | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT                                                                     |      |            |             |                                    |          |               |                        |                                                             |                        | -BGI24, -BGI25            | POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -BGA2               | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM) |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   | F     |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BGA3               | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT                                                                               |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BGA4               | POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)                                                  |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BGB1....-BGB3      | POSITION SWITCHES OF CIRCUIT-BREAKER                                                                                                                |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -BGB4               | POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS                                                      |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| <table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td colspan="3">PROJECT TITLE</td><td rowspan="3"><b>ABB</b> <b>EPMV</b></td><td rowspan="3">REFERENCE DESIGNATIONS<br/>MVD-20 &amp; MVD-21 20kV Boiler Plant</td><td>Scale</td><td>NTS</td><td>= H05</td></tr><tr><td></td><td></td><td></td><td></td><td>Prepared</td><td>YR</td><td colspan="3" rowspan="2">MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>&amp; 307</td></tr><tr><td>00</td><td>Issued for Approval</td><td>11.01.19</td><td>YR</td><td>Checked</td><td>AW</td><td>PT. ABB Sakti Industri</td><td>SCHEMATIC DIAGRAM</td><td>Drawing Nbr.</td><td>18.042-MNA-307.03</td><td>Sh. 3</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td>SBS-W BUS TIE</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 27</td></tr></table> |                     |                                                                                                                                                     |      |            |             |                                    |          |               |                        |                                                             |                        |                           |                                                                                                                                     |                   |       |   |  |   | Date Prep. | 15.11.2018 | PROJECT TITLE |  |  | <b>ABB</b> <b>EPMV</b> | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant | Scale | NTS | = H05 |  |  |  |  | Prepared | YR | MV Switchgear for MNA Serang Plant |  |  | Customer | PT Multimas Nabati Asahan | & 307 | 00 | Issued for Approval | 11.01.19 | YR | Checked | AW | PT. ABB Sakti Industri | SCHEMATIC DIAGRAM | Drawing Nbr. | 18.042-MNA-307.03 | Sh. 3 | Ind. | Revision | Date | Name | Approved | AI | Based on: | BE318042 | SBS-W BUS TIE | Customer Doc. Nbr. |  | Cont. 27 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                                     |      | Date Prep. | 15.11.2018  | PROJECT TITLE                      |          |               | <b>ABB</b> <b>EPMV</b> | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant | Scale                  | NTS                       | = H05                                                                                                                               |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                                     |      | Prepared   | YR          | MV Switchgear for MNA Serang Plant |          |               |                        |                                                             | Customer               | PT Multimas Nabati Asahan | & 307                                                                                                                               |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Issued for Approval | 11.01.19                                                                                                                                            | YR   | Checked    | AW          |                                    |          |               |                        |                                                             | PT. ABB Sakti Industri | SCHEMATIC DIAGRAM         | Drawing Nbr.                                                                                                                        | 18.042-MNA-307.03 | Sh. 3 |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| Ind.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Revision            | Date                                                                                                                                                | Name | Approved   | AI          | Based on:                          | BE318042 | SBS-W BUS TIE | Customer Doc. Nbr.     |                                                             | Cont. 27               |                           |                                                                                                                                     |                   |       |   |  |   |            |            |               |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |                                                                                                                                                     | 2    |            |             | 3                                  |          |               | 4                      |                                                             |                        | 5                         |                                                                                                                                     |                   | 6     |   |  | 7 |            |            | 8             |  |  |                        |                                                             |       |     |       |  |  |  |  |          |    |                                    |  |  |          |                           |       |    |                     |          |    |         |    |                        |                   |              |                   |       |      |          |      |      |          |    |           |          |               |                    |  |          |





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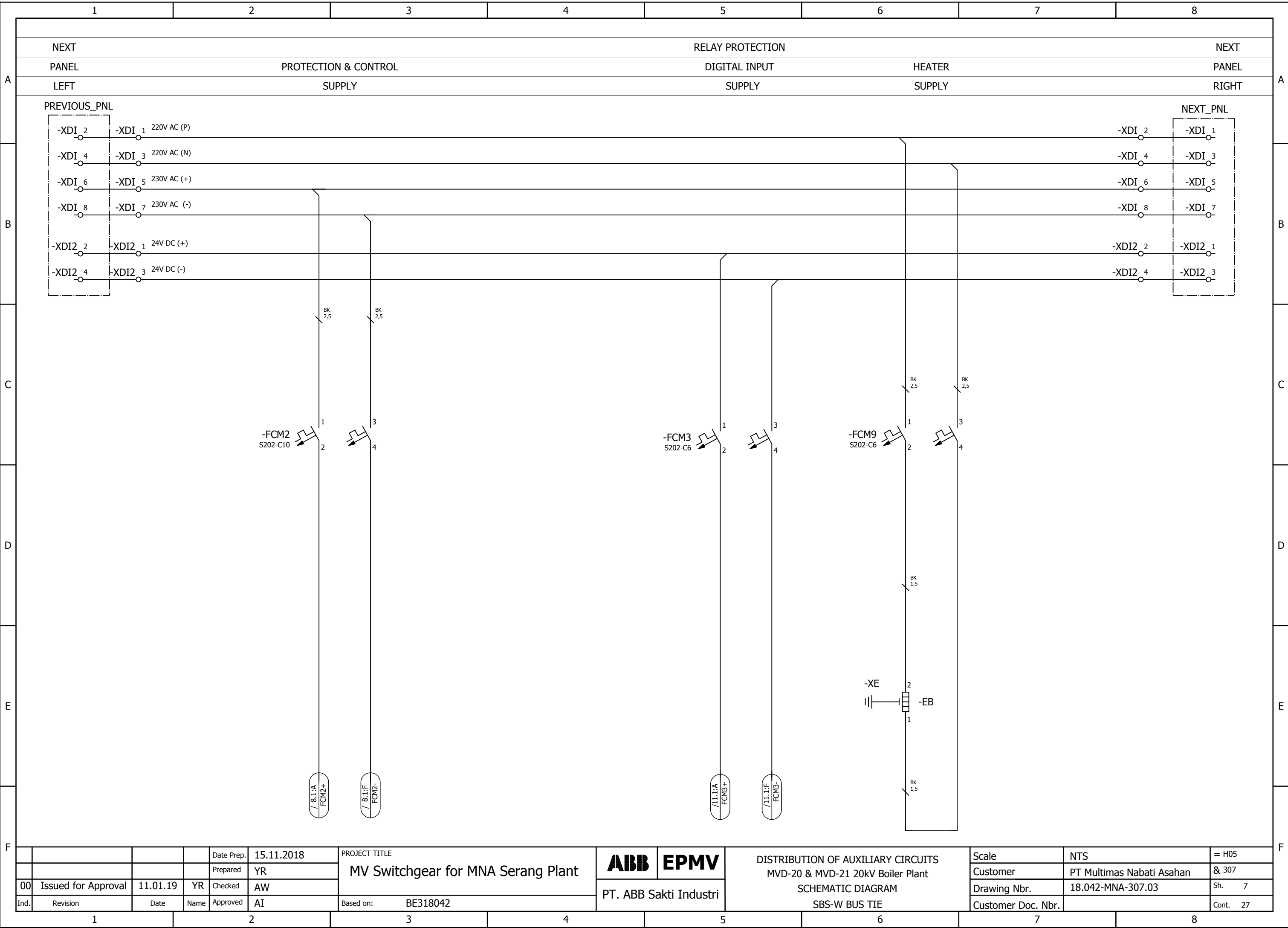
|   |                 |                                                                                                    |       |                                                                                               |
|---|-----------------|----------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------|
| A | -TFA            | ACTIVE POWER TRANSDUCERS                                                                           | -XDI1 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS                                       |
|   | -TFC            | CURRENT TRANSDUCERS                                                                                | -XDI2 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING |
|   | -TFF            | FREQUENCY TRANSDUCERS                                                                              | -XDI3 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION    |
|   | -TFJ            | ACTIVE ENERGY TRANSDUCERS                                                                          | -XDI4 | TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS                                |
|   | -TFK            | REACTIVE ENERGY TRANSDUCERS                                                                        | -XDI6 | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS            |
|   | -TFM            | MULTIFUNCTION TRANSDUCERS                                                                          | -XDI7 | TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS                                                 |
|   | -TFP            | POWER-FACTOR TRANSDUCERS                                                                           | -XDI8 | TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS                                       |
|   | -TFQ            | REACTIVE POWER TRANSDUCERS                                                                         | -XDI9 | TERMINAL BLOCK FOR SPECIAL USE                                                                |
|   | -TFS            | SIGNAL CONVERTERS                                                                                  | -XDJ  | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD                                   |
|   | -TFV            | VOLTAGE TRANSDUCERS                                                                                | -XDJ1 | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS                             |
| B | -WA             | BUSBARS                                                                                            | -XDM  | SEALABLE TERMINAL BLOCK FOR MEASUREMENT                                                       |
|   | -WBC            | POWER CABLES                                                                                       | -XDS  | CURRENT SOCKETS                                                                               |
|   | -WE             | EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.                                                   | -XDT  | TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB                           |
|   | -WGC            | CONTROL CABLES                                                                                     | -XDU  | CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                        |
|   | -WGS            | COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)                                                      | -XDU1 | TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK                                |
|   | -WH             | OPTICAL FIBRES                                                                                     | -XDV  | TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS                                           |
|   | -XDA            | TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS                                                | -XDV1 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1                            |
|   | -XDA1           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1                                 | -XDV2 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2                            |
|   | -XDA2           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2                                 | -XDV3 | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3                            |
|   | -XDA3           | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3                                 | -XDV4 | CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR                                         |
| C | -XDA5           | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDX  | SUPPORT TERMINAL BLOCK                                                                        |
|   | -XDB            | CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS | -XE1  | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE                                        |
|   | -XDB1           | CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)           | -XE2  | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE                                       |
|   | -XDB2           | CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER                                     | -XE5  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE                           |
|   | -XDB10...-XDB89 | CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS                                                    | -XE6  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE                                 |
|   | -XDB7           | CONNECTOR FOR TRUCK POSITION CONTACTS                                                              | -XE7  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE                                        |
|   | -XDB91          | CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE8  | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE                                |
|   | -XDB92          | CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      | -XE9  | EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT                                |
|   | -XDB93          | CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR                                             |       |                                                                                               |
|   | -XDB95          | CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET                                           |       |                                                                                               |
| D | -XDC            | CUSTOMER TERMINAL BLOCK                                                                            |       |                                                                                               |
|   | -XDC1           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER                                  |       |                                                                                               |
|   | -XDC2           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR                              |       |                                                                                               |
|   | -XDC3           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT          |       |                                                                                               |
|   | -XDC4           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS                             |       |                                                                                               |
|   | -XDC5           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER                        |       |                                                                                               |
|   | -XDE            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE                                      |       |                                                                                               |
|   | -XDE1           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE                                       |       |                                                                                               |
|   | -XDE2           | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                            |       |                                                                                               |
|   | -XDE3           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                             |       |                                                                                               |
| E | -XDF            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS                                |       |                                                                                               |
|   | -XDF1           | CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR                                            |       |                                                                                               |
|   | -XDF2           | CONNECTOR FOR REAR VENTILATOR CIRCUITS                                                             |       |                                                                                               |
|   | -XDF3           | CONNECTOR FOR FRONT VENTILATOR CIRCUITS                                                            |       |                                                                                               |
|   | -XDH            | TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)   |       |                                                                                               |
|   | -XDI            | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)                                     |       |                                                                                               |

|      |                     |          |      |            |            |                                                     |          |                        |      |                                                             |  |                    |                           |          |
|------|---------------------|----------|------|------------|------------|-----------------------------------------------------|----------|------------------------|------|-------------------------------------------------------------|--|--------------------|---------------------------|----------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          | ABB                    | EPMV | REFERENCE DESIGNATIONS<br>MVD-20 & MVD-21 20kV Boiler Plant |  | Scale              | NTS                       | = H05    |
|      |                     |          |      | Prepared   | YR         |                                                     |          |                        |      |                                                             |  | Customer           | PT Multimas Nabati Asahan | & 307    |
| 00   | Issued for Approval | 11.01.19 | YR   | Checked    | AW         |                                                     |          |                        |      |                                                             |  | Drawing Nbr.       | 18.042-MNA-307.03         | Sh. 6    |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                           | BE318042 | PT. ABB Sakti Industri |      | SCHEMATIC DIAGRAM<br>SBS-W BUS TIE                          |  | Customer Doc. Nbr. |                           | Cont. 27 |

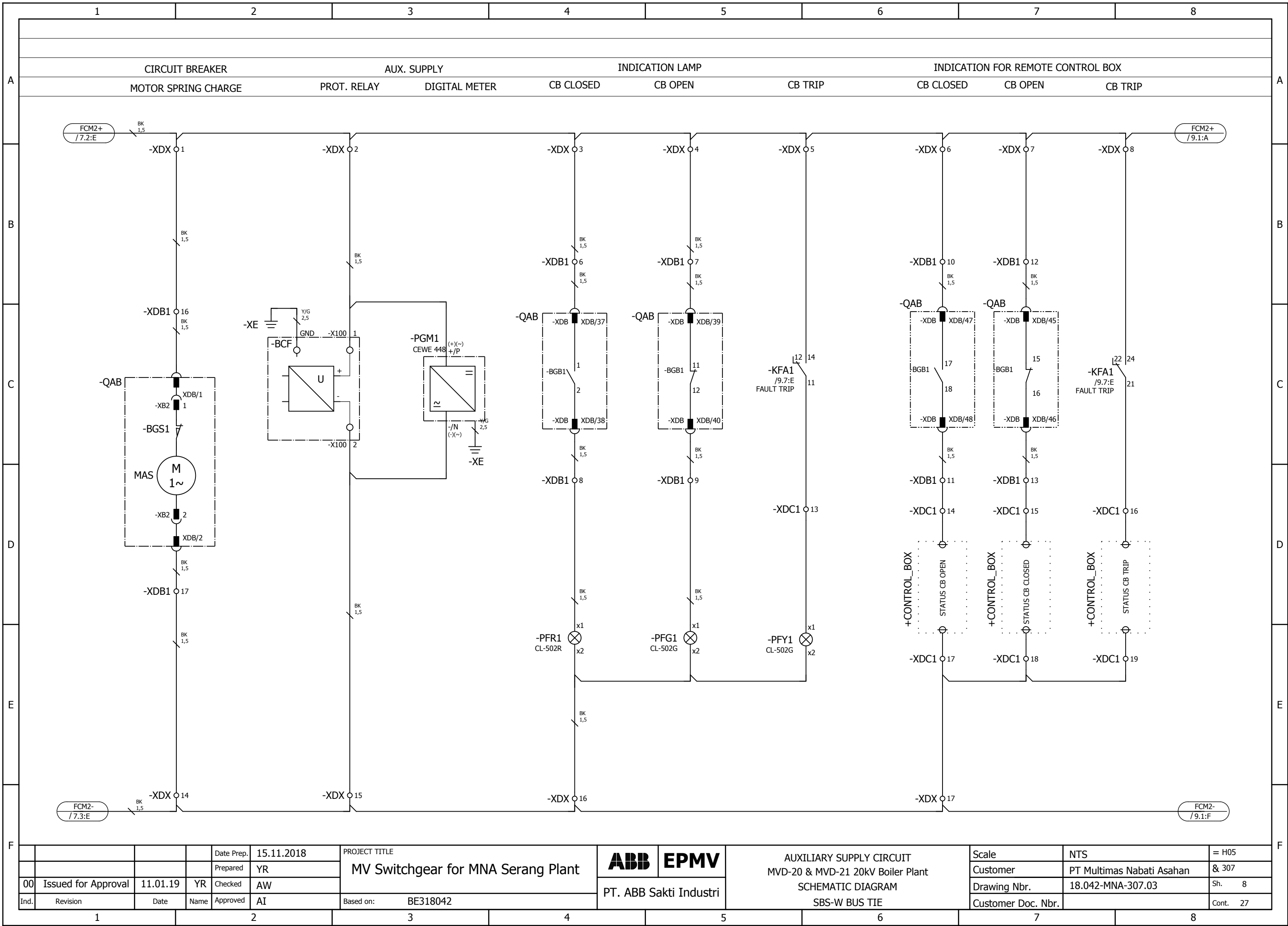
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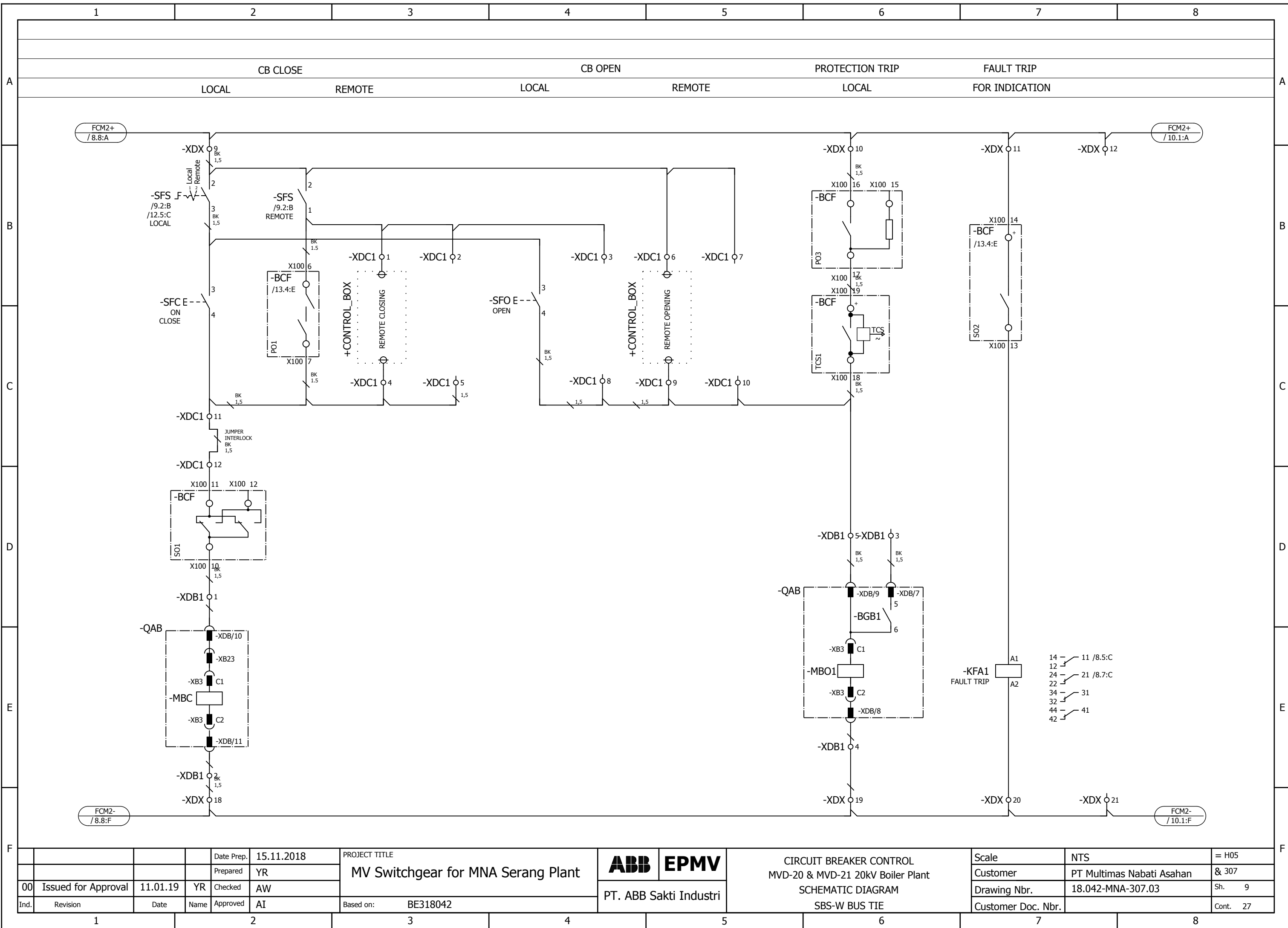


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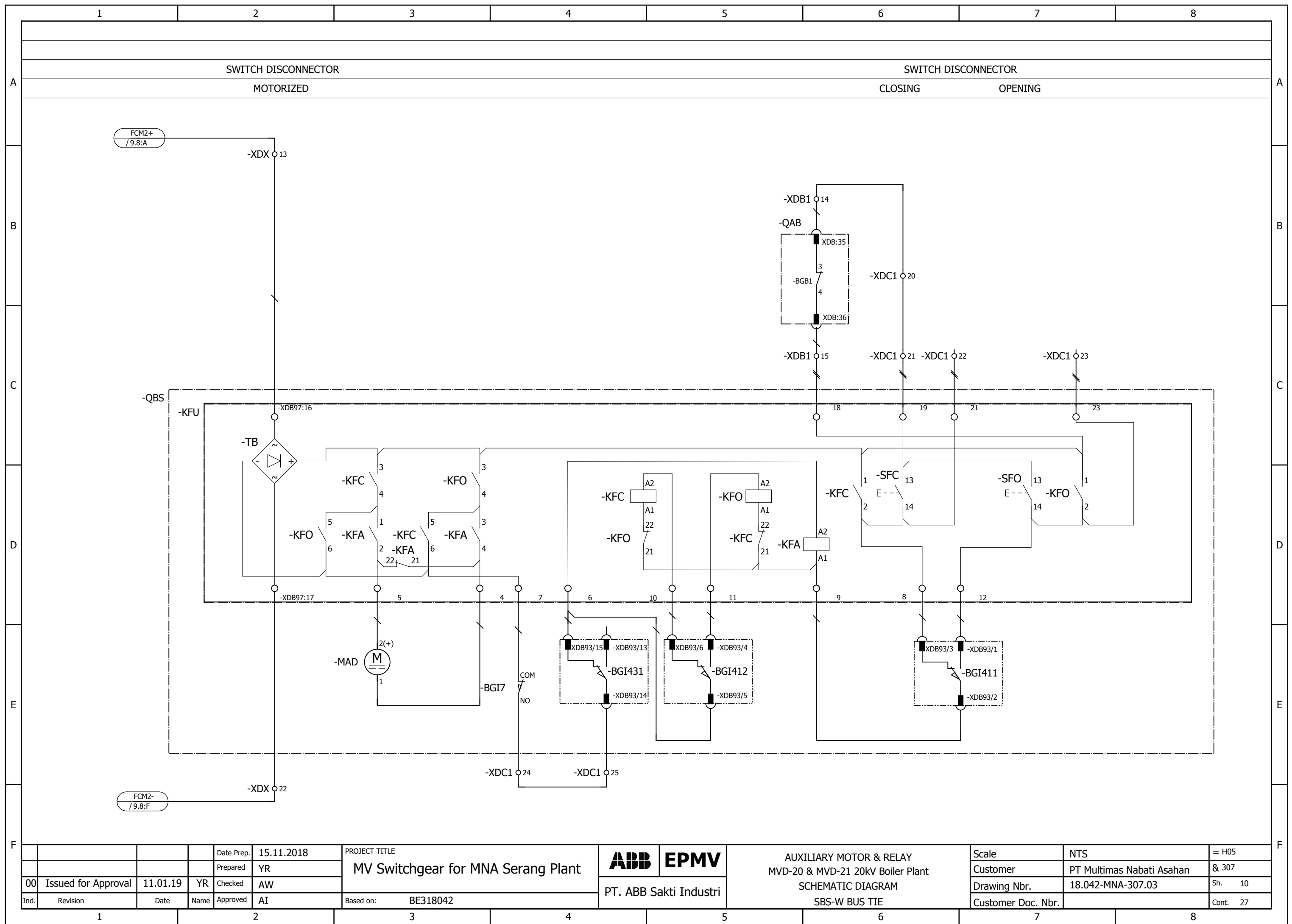


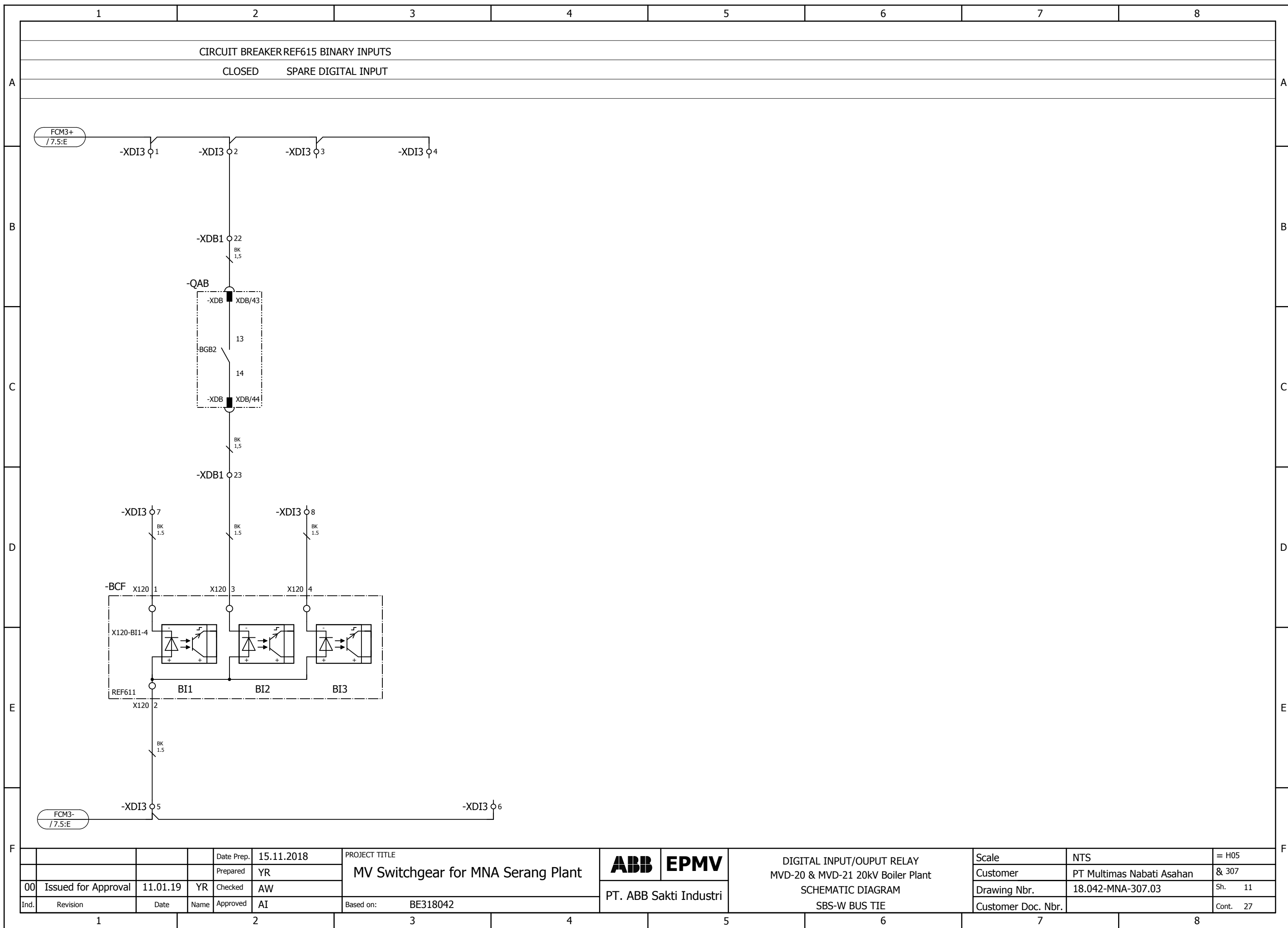


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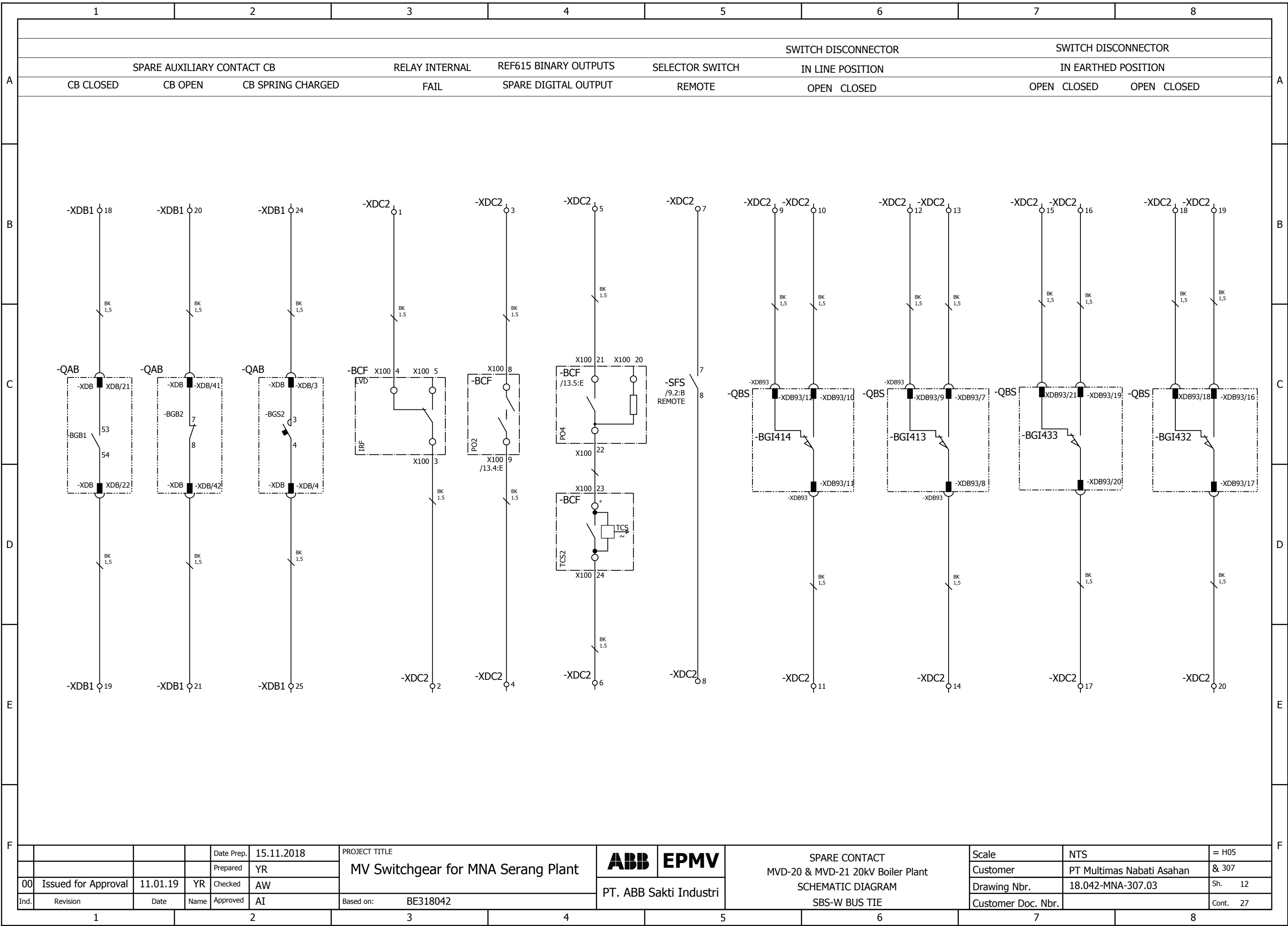


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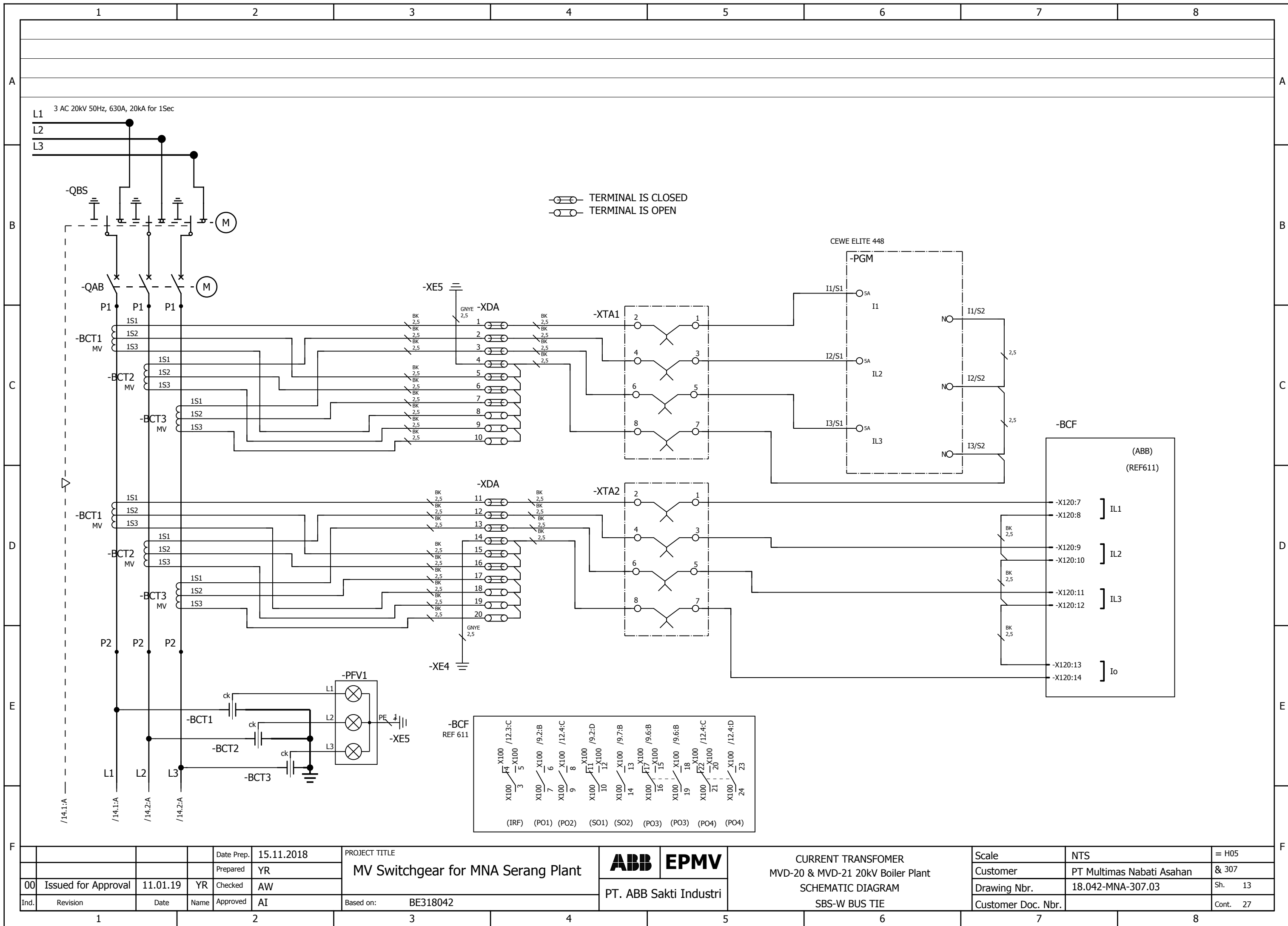


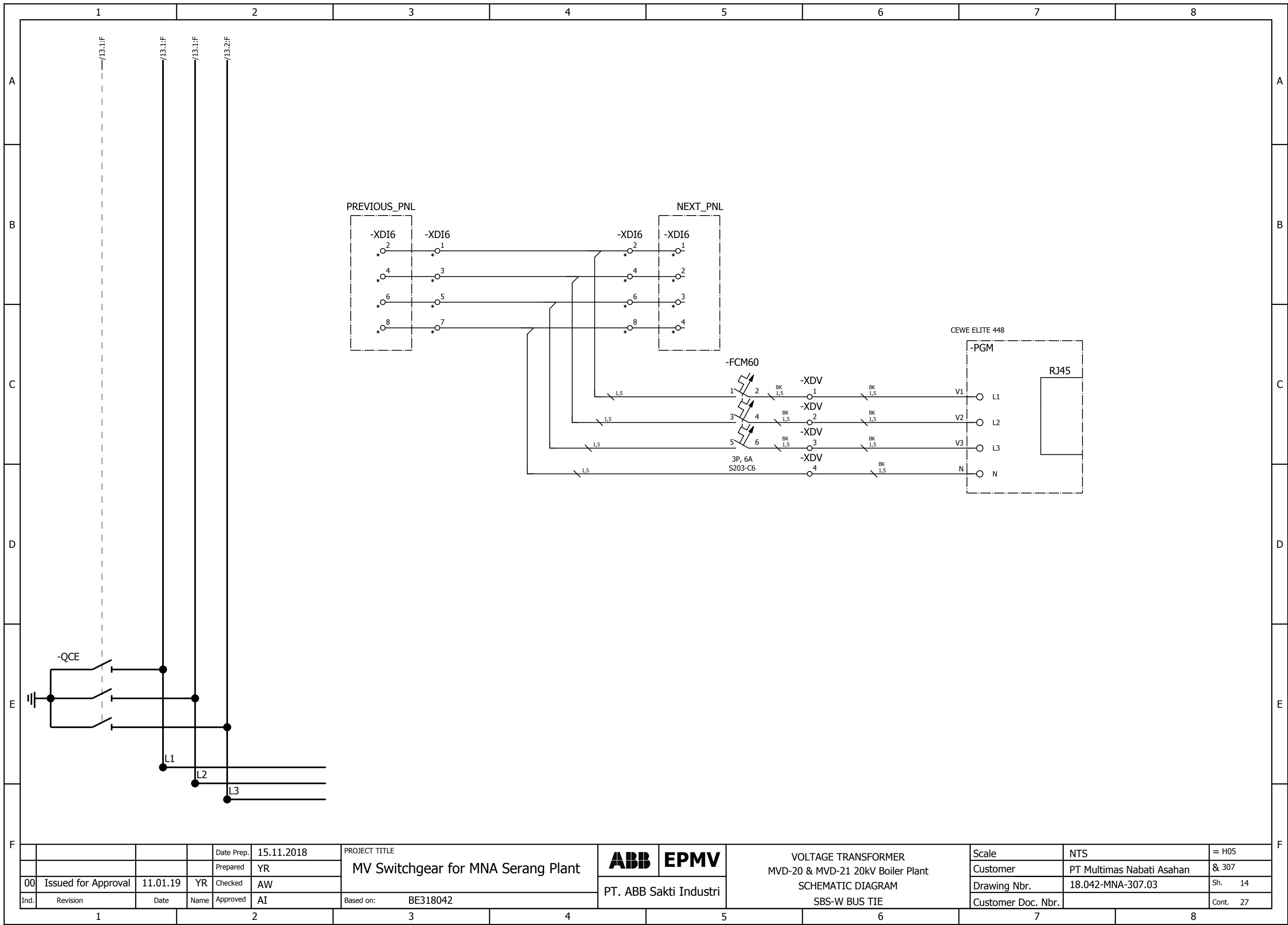


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| No. | EQUIPMENT DESCRIPTION                            | ENGRAVING                 |
|-----|--------------------------------------------------|---------------------------|
| 1   | -SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)     | -SFS, LOCAL-OFF-REMOTE    |
| 2   | -SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)          | -SFC, CB CLOSE            |
| 3   | -SFO (PUSH BUTTON CB OPEN, RED, ABB)             | -SFO, CB OPEN             |
| 4   | -PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB) | -PFY1, CB TRIP            |
| 5   | -PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)   | -PFR1, CB CLOSE           |
| 6   | -PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)  | -PFG1, CB OPEN            |
| 7   | -PGJ (DIGITAL MEASUREMENT, CEWE 448)             | -PGJ, DIGITAL MEASUREMENT |
| 8   | -BCF (FEEDER PROTECTION RELAY, REF615J, ABB)     | -BCF, RELAY PROTECTION    |
| 9   | -XTA1, XTA2 (TEST BLOCK , YONGSUNG)              | -XTA, CURRENT TEST BLOCK  |
|     |                                                  |                           |
|     |                                                  |                           |
|     |                                                  |                           |
|     |                                                  |                           |

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| 1                     | 2                   | 3        | 4                                                                                                               | 5          | 6                                  | 7                      | 8                   |                 |                                   |               |              |                           |        |          |
|-----------------------|---------------------|----------|-----------------------------------------------------------------------------------------------------------------|------------|------------------------------------|------------------------|---------------------|-----------------|-----------------------------------|---------------|--------------|---------------------------|--------|----------|
| Summarized parts list |                     |          |                                                                                                                 |            |                                    |                        | IDABB_F02_005-IDABB |                 |                                   |               |              |                           |        |          |
| Carry over: 0.00      |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
| A                     | TAG ID              |          | DESCRIPTION                                                                                                     |            | MANUFACTUR/BRAND                   | PART NUMBER            | QUANTITY            | REMARKS         | A                                 |               |              |                           |        |          |
| B                     | -BCF                |          | Feeder Management Relay<br>REF611HBAABC1NN11G                                                                   |            | ABB                                | REF611HBAABC1NN11G     | 1                   | /8.2:C          | B                                 |               |              |                           |        |          |
|                       | -EB                 |          | HEATER<br>230VAC, 50W                                                                                           |            | SANYANG                            | CDXN010033P0009        | 1                   | /7.6:E          |                                   |               |              |                           |        |          |
|                       | -FCM2               |          | MCB S202-C10<br>MCB. 2P, 10A                                                                                    |            | ABB                                | 2CDS252001R0104        | 1                   | /7.2:C          |                                   |               |              |                           |        |          |
|                       | -FCM3;-FCM9         |          | MCB S202-C6<br>MCB. 2P, 6A                                                                                      |            | ABB                                | 2CDS252001R0064        | 2                   | /7.5:C;/7.6:C   |                                   |               |              |                           |        |          |
|                       | -FCM60              |          | MCB S203-C6<br>MCB. 3P, 6A                                                                                      |            | ABB                                | 2CDS253001R0064        | 1                   | /14.5:C         |                                   |               |              |                           |        |          |
| C                     | -KFA1               |          | Auxiliary Relay<br>230AC 4COS                                                                                   |            | PHOENIX                            | 1YRA3DL266230001       | 1                   | /9.7:E          | C                                 |               |              |                           |        |          |
|                       | -PFG1               |          | PILOT LIGHT GREEN #CL-523G#230VAC<br>CL-523G                                                                    |            | ABB                                | 1SFA619402R5232        | 1                   | /8.5:E          |                                   |               |              |                           |        |          |
|                       | -PFR1               |          | PILOT LIGHT RED #CL-523R#230VAC<br>CL-523R                                                                      |            | ABB                                | 1SFA619402R5231        | 1                   | /8.4:E          |                                   |               |              |                           |        |          |
|                       | -PFY1               |          | PILOT LIGHT YELLOW #CL-523Y#230VAC<br>CL-523Y                                                                   |            | ABB                                | 1SFA619402R5233        | 1                   | /8.6:E          |                                   |               |              |                           |        |          |
| D                     | -PGM1               |          | POWER METER CEWE 448<br>Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC     |            | CEWE                               | 1YRA4EA0350001         | 1                   | /8.3:C          | D                                 |               |              |                           |        |          |
|                       | -SFC                |          | PUSH BUTTON, YW1B-M1E10-G<br>GREEN COLOUR                                                                       |            | IDEC                               | 1YRA3DA14700001        | 1                   | /9.2:B          |                                   |               |              |                           |        |          |
|                       | -SFO                |          | PUSH BUTTON, YW1B-M1E10-R<br>RED COLOUR                                                                         |            | IDEC                               | 1YRA3DA14700002        | 1                   | /9.4:B          |                                   |               |              |                           |        |          |
|                       | -SFS                |          | LOCAL/REMOTE Selector Switch<br>Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position) |            | K&N                                | 1YRA3DA17940011        | 1                   | /9.2:B          |                                   |               |              |                           |        |          |
| E                     | -XTA1;-XTA2         |          | Current Test Block<br>Rated Current : 10A, 250V; Mounting : Flush Mounting                                      |            | YONGSUNG                           | YSCTT-04C              | 2                   | /13.4:C;/13.4:D | E                                 |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
| F                     |                     |          |                                                                                                                 |            |                                    |                        |                     |                 | F                                 |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          |                                                                                                                 |            |                                    |                        |                     |                 |                                   |               |              |                           |        |          |
|                       |                     |          | Date Prep.                                                                                                      | 15.11.2018 | PROJECT TITLE                      |                        | ABB                 | EPMV            | PART LIST                         |               | Scale        | NTS                       | = H05  |          |
|                       |                     |          | Prepared                                                                                                        | YR         | MV Switchgear for MNA Serang Plant |                        |                     |                 | MVD-20 & MVD-21 20kV Boiler Plant |               | Customer     | PT Multimas Nabati Asahan | & 307  |          |
| 00                    | Issued for Approval | 11.01.19 | YR                                                                                                              | Checked    | AW                                 | PT. ABB Sakti Industri |                     |                 | SCHEMATIC DIAGRAM                 |               | Drawing Nbr. | 18.042-MNA-307.03         | Sh. 17 |          |
| Ind.                  | Revision            | Date     | Name                                                                                                            | Approved   | AI                                 |                        |                     |                 | Based on:                         | SBS-W BUS TIE |              | Customer Doc. Nbr.        |        | Cont. 27 |
| 1                     |                     |          | 2                                                                                                               |            |                                    | 3                      | 4                   | 5               | 6                                 | 7             | 8            |                           |        |          |

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| F                                                                                                                                                                       |  | E          |  | D          |           | C                                  |                 | B        |          | A                  |       |                           |          |   |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------|--|------------|-----------|------------------------------------|-----------------|----------|----------|--------------------|-------|---------------------------|----------|---|--|
| <div>Ind.</div> <div>Revision</div> <div>Date</div> <div>Name</div> <div>Approved</div>                                                                                 |  | REMARK     |  | CABLE      | EQUIPMENT | Wire No                            | Terminal number | TYPE     | WIRE NO. | EQUIPMENT          | CABLE | REMARK                    |          |   |  |
|                                                                                                                                                                         |  | BK 1,5 mm² |  |            | -PGM:V1   |                                    | 1               | PT 2.5   |          | -FCM60:2           |       | BK 1,5 mm²                |          |   |  |
|                                                                                                                                                                         |  | BK 1,5 mm² |  |            | -PGM:V2   |                                    | 2               | PT 2.5   |          | -FCM60:4           |       | BK 1,5 mm²                |          |   |  |
|                                                                                                                                                                         |  | BK 1,5 mm² |  |            | -PGM:V3   |                                    | 3               | PT 2.5   |          | -FCM60:6           |       | BK 1,5 mm²                |          |   |  |
|                                                                                                                                                                         |  | BK 1,5 mm² |  |            | -PGM:N    |                                    | 4               | PT 2.5   |          | -XDI6:8            |       | 1,5 mm²                   |          |   |  |
| ==MVD-20_MVD-21=H05+--XDV                                                                                                                                               |  |            |  |            |           |                                    |                 |          |          |                    |       |                           |          |   |  |
| REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS.<br>△ OUTGOING CONTROL CABLE (Supply by Customer).<br>● BRIDGE BAR.<br>✕ PARTITION PLATE.<br>( TEST JUMPER. |  |            |  |            |           |                                    |                 |          |          |                    |       |                           |          |   |  |
| PROJECT TITLE                                                                                                                                                           |  | Date Prep. |  | 15.11.2018 |           | MV Switchgear for MNA Serang Plant |                 | ABB EPMV |          | Terminal diagram   |       | Scale                     | = H05    |   |  |
| Based on:                                                                                                                                                               |  | Prepared   |  | YR         |           | PT. ABB Sakti Industri             |                 |          |          | Customer           |       | PT Multimas Nabati Asahan | & 307    |   |  |
|                                                                                                                                                                         |  | Checked    |  | AW         |           |                                    |                 |          |          | Drawing Nbr.       |       |                           | Sh. 26   |   |  |
|                                                                                                                                                                         |  | Approved   |  | AI         |           |                                    |                 |          |          | Customer Doc. Nbr. |       |                           | Cont. 27 |   |  |
| 1                                                                                                                                                                       |  | 2          |  | 3          |           | 4                                  |                 | 5        |          | 6                  |       | 7                         |          | 8 |  |

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